Our goal for this project is to investigate whether artificial neural networks process images similar to human memory systems. We will use Representational Similarity Analysis as a metric to gauge a comparison between how pretrained convolutional neural networks encode images in relation to human memorability judgements. This will be done through extracting high-dimensional feature representations from several pre-trained deep learning networks such as ResNet and compute pairwise similarity measures to construct representational dissimilarity matrices (RDM's) and compare them to the human memorability scores in the LaMem dataset. We will assess the alignment between these scores and investigate what visual or representative features may explain any differences.

ResNet
DensNet
EfficientNet
MemNet
ResMemNet

Novel CNN combinations you could explore

Below are combinations of CNNs (or families) that, to the best of current literature, have not been systematically compared as a *group* on human memorability. These would be strong candidates for novel work, especially if evaluated on LaMem, Isola's dataset, MemCat, and face-memorability datasets.

1. Modern efficient CNNs vs classic backbones

Group design (all fine-tuned for memorability):

- Classic: AlexNet, VGG16/19, ResNet-50 (baseline group widely used).
- Efficient: MobileNetV2/V3, EfficientNet-B0/B4, DenseNet-121/169.
- Hypothesis: Efficient models may match or exceed memorability prediction with far fewer parameters; you can explicitly trade off accuracy vs efficiency, something not reported in memorability work.

Novelty: A systematic "classic vs efficient" comparison with human consistency benchmarks is, as far as current surveys indicate, missing.

2. Family-wise scaling within one architecture

Choose one family and examine memorability behavior across depth/width:

- ResNet: 18, 34, 50, 101, 152.
- EfficientNet: B0–B7.
- RegNet or ResNeXt family: multiple capacity points.

Novelty: Memorability work typically uses a single ResNet backbone (often 50) rather than systematic scaling; showing how memorability prediction approaches human consistency as capacity increases would be new.

3. Lightweight vs heavyweight for mobile memorability applications

Group:

- Lightweight: MobileNetV3-Small, ShuffleNetV2, SqueezeNet, MNASNet.
- Heavyweight: VGG16, ResNet-152, Inception-ResNet-v2.

Novelty: No existing paper appears to target "on-device memorability prediction" with a rigorous accuracy–latency–energy trade-off across multiple CNNs.

4. Cross-domain pretraining and hybrid features

Existing work has exploited ImageNet + Places hybrid models (e.g., VGG hybrid) and ResNet backbones, but usually in limited forms (e.g., MemNet, ResMem, IncResMem). New combinations:

- Scene-centric vs object-centric:
    - Compare memorability predictors built from:
        - Places-only CNNs (e.g., Places365-ResNet).
        - ImageNet-only CNNs (ResNet-50/101).
        - Hybrid ImageNet+Places backbones (like the hybrid VGG used in IncResMem).
- Multi-backbone ensembles:

- Combine DenseNet + EfficientNet + ResNet features in a simple learned fusion head for memorability.

Novelty: Scene/object pretraining effects are hinted at but not systematically explored across many backbones.

5. Architectures rarely or not used in memorability

You could create novel groups explicitly from underused families (as backbones for a common memorability head):

- Group A: DenseNet-121, DenseNet-201.
- Group B: EfficientNet-B0, EfficientNet-B4, EfficientNet-V2-S/M.
- Group C: RegNetY-X (several capacity points) and ResNeXt-50/101.
- Group D: Vision-style CNNs optimized via NAS for classification (e.g., NASNet-Mobile/Large) repurposed for memorability.

As far as current surveys, these specific families are not discussed as core memorability models, making a direct, same-protocol comparison genuinely new.

How to frame a "no one did this" study

To make a strong novelty claim, you would:

1. Use a standard dataset (LaMem, Isola et al., MemCat, face memorability datasets) and standard metrics (Spearman ρ vs human scores, comparison to human-human consistency).
2. Implement a *uniform* memorability head (e.g., global pooling + small MLP regression) on top of each pretrained backbone (to control for head differences).
3. Define explicit groups (e.g., classic vs efficient vs lightweight vs scaled-capacity) and compare their performance and "memorability saliency maps" (Grad-CAM/GradRAM).
4. Show where each group reaches or fails to reach human consistency, and analyze which visual properties each group emphasizes (e.g., object vs scene vs texture).

- MobileNet (V1/V2/V3).
- DenseNet family.
- NASNet (large/mobile), EfficientNet, EfficientNetV2.
- RegNet, ResNeXt, Wide-ResNet variants (beyond ResNet-50 style backbones used in ResMem-type work).
- SqueezeNet, ShuffleNet, MNASNet and other lightweight architectures